Homework 4: Test for Stutter

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Course: Secure Software Testing

Program File	Stutter.js
GitHub Repository	https://github.com/tainyu/secure-software-testing-2025-spring-classroom-lab-2-114-spring-secure-software-testing-1/tree/main/HW4

This report answers parts (a)-(d) of Homework 4 based on the supplied Stutter.js program and follows the submission guideline requiring a GitHub link and named report file.

(a) Control Flow Graphs

The following CFGs are drawn for all functions defined in Stutter.js: stut(inFile), checkDupes(line), isDelimit(C), and main().

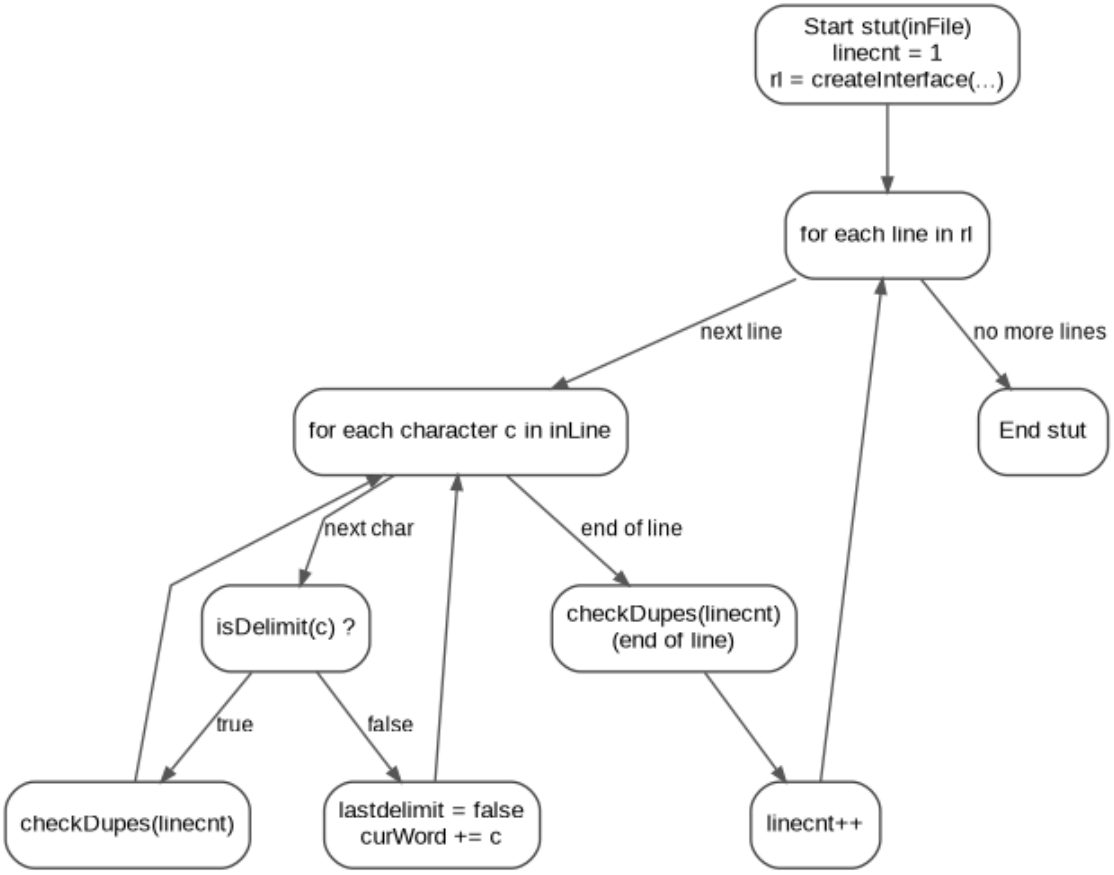

Figure 1. CFG for stut(inFile)

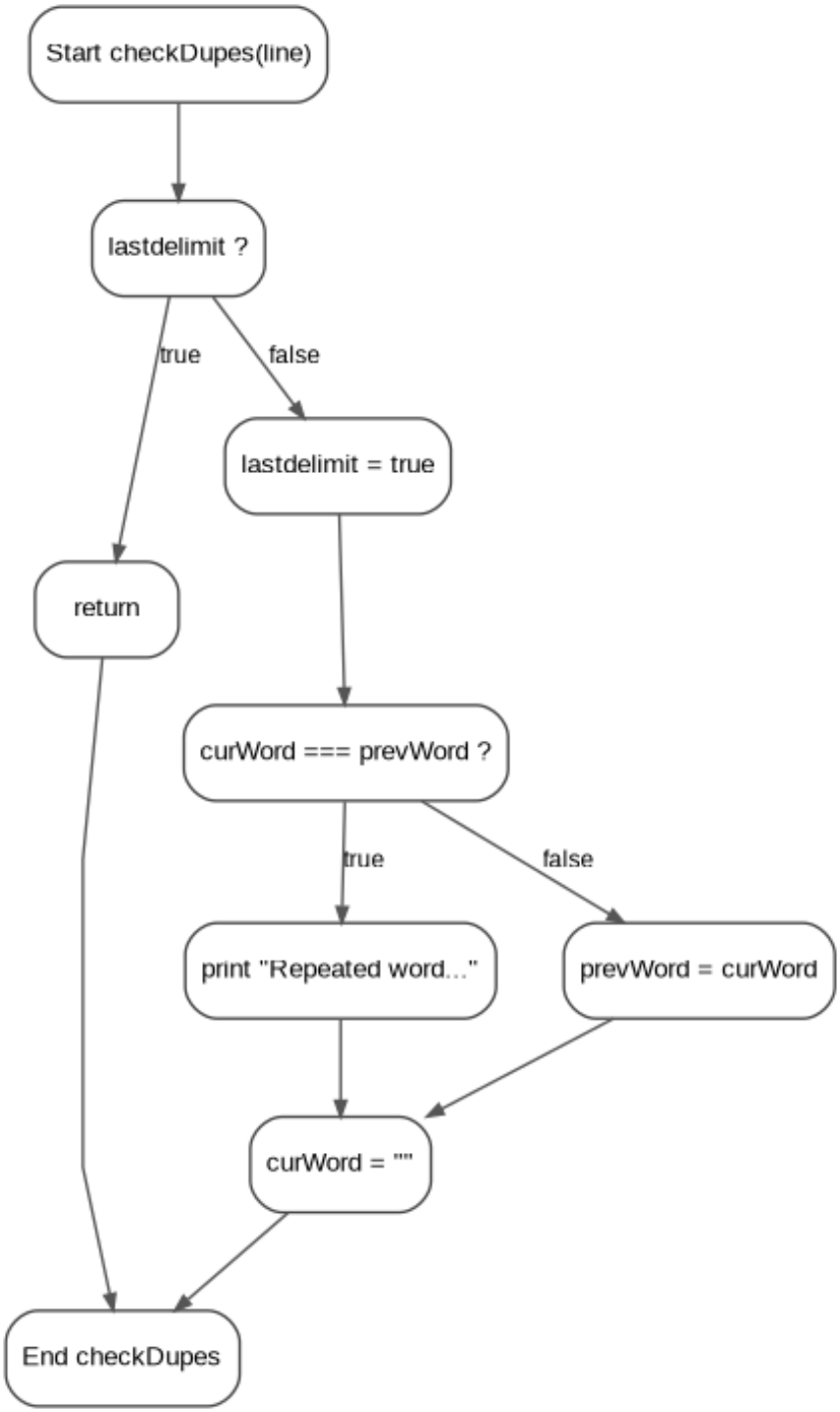

Figure 2. CFG for checkDupes(line)

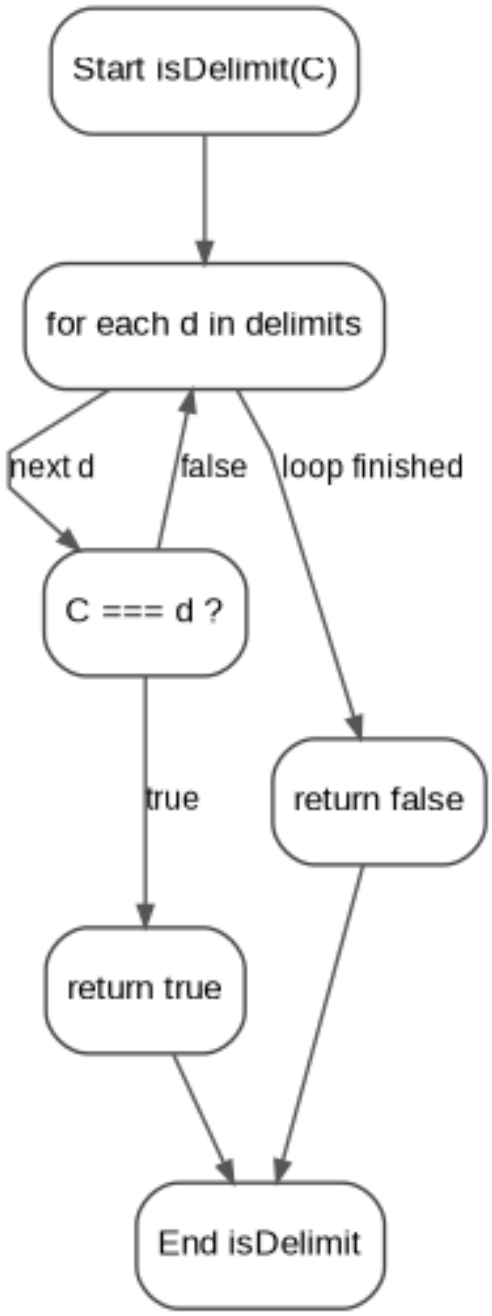

Figure 3. CFG for isDelimit(C)

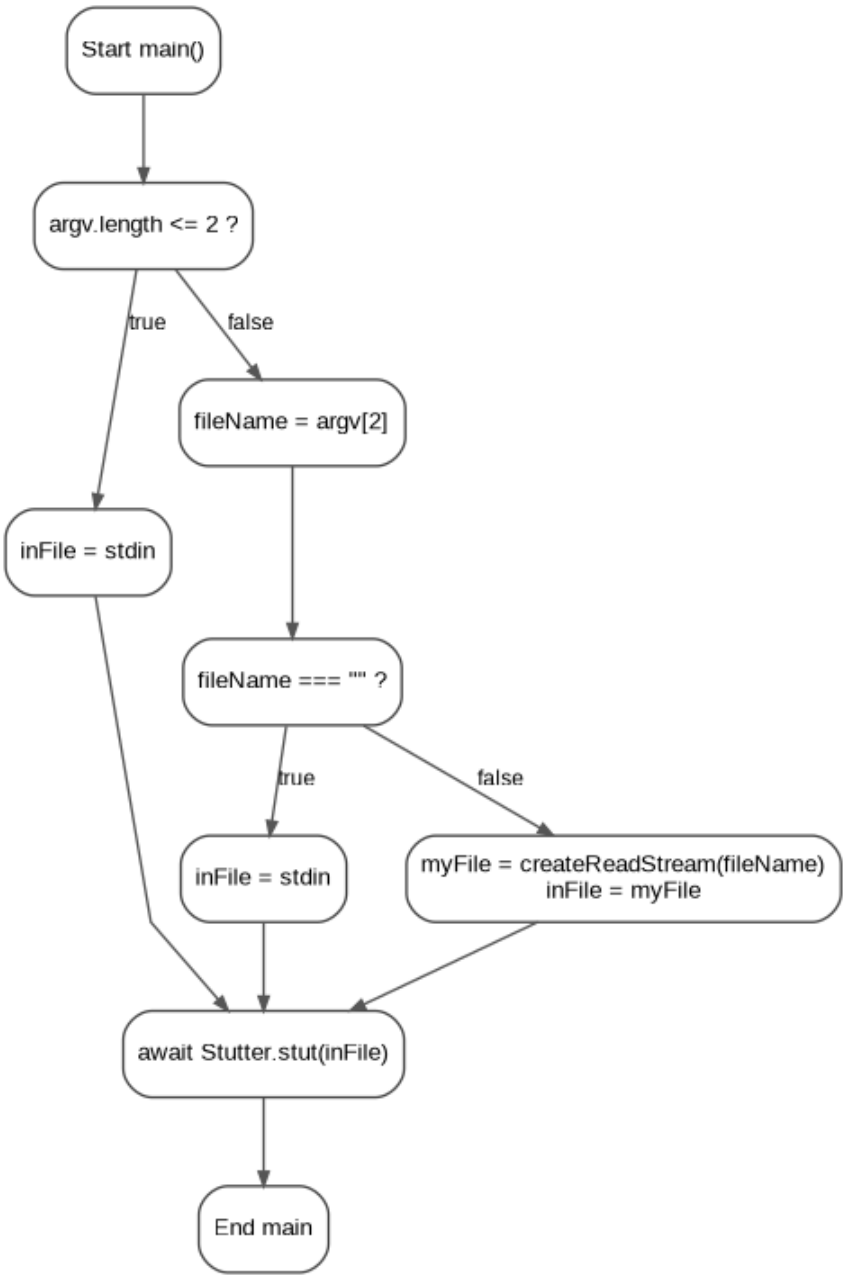

Figure 4. CFG for main()

(b) Call Sites of Functions Defined in Stutter.js

The table below lists every call site whose callee is a function defined inside Stutter.js. Calls to library functions such as fs.createReadStream() are excluded because they are not functions defined in Stutter.js.

No.	Caller	Call Site	Callee	Notes
1	stut(inFile)	if (this.#isDelimit(c))	isDelimit(C)	Called once for each character in a line.
2	stut(inFile)	this.#checkDupes(linecnt) inside delimiter branch	checkDupes(line)	Triggered when a delimiter ends a word before end-of-line.
3	stut(inFile)	this.#checkDupes(linecnt) after inner loop	checkDupes(line)	Triggered at end of each input line.
4	main()	await Stutter.stut(inFile)	stut(inFile)	The selected input stream is passed to stut().
5	module entry	if (require.main === module) { main(); }	main()	Program-start call site; no actual parameters are passed.

(c) Coupling DU-Pairs for Each Call Site in (b)

A coupling DU-pair connects an actual parameter definition in the caller to a reachable use of the corresponding formal parameter in the callee. For the module-entry call to main(), there is no parameter passing, so no coupling DU-pair exists.

No.	Call Site	Actual-In Def	Formal-In	Reachable Use in Callee	Result
1	stut() -> isDelimit(c)	c defined by: for (const c of inLine)	C	Used in decision: if (C === d)	DU-pair exists
2	stut() -> checkDupes(linecnt) [delimiter call site]	linecnt defined by initialization linecnt = 1 or by linecnt++ from earlier iterations	line	Used in output: Repeated word on line ${line} ...	DU-pair exists
3	stut() -> checkDupes(linecnt) [end-of-line call site]	linecnt defined by initialization linecnt = 1 or by linecnt++ from earlier iterations	line	Used in output: Repeated word on line ${line} ...	DU-pair exists
4	main() -> stut(inFile) [argv.length <= 2 path]	inFile = stdin	inFile	Used in readline.createInterface({ input: inFile, ... })	DU-pair exists
5	main() -> stut(inFile) [fileName === "" path]	inFile = stdin	inFile	Used in readline.createInterface({ input: inFile, ... })	DU-pair exists

No.	Call Site	Actual-In Def	Formal-In	Reachable Use in Callee	Result
6	main() -> stut(inFile) [valid file path]	myFile = fs.createReadStream(fileName); inFile = myFile	inFile	Used in readline.createInterface({ input: inFile, ... })	DU-pair exists
7	module entry -> main()	No actual parameter	None	None	No coupling DU-pair

(d) Test Data for All-Coupling-Uses Coverage

The following test set covers all coupling uses identified above. The key idea is to force each actual-parameter definition to reach at least one real use in the callee. For checkDupes(line), the formal parameter line is only used when a repeated word is reported, so the test data must actually trigger the duplicate-word message.

Test	Execution Mode	Input / File Content	Coverage Purpose	Expected Result	AUC Covered
T1	node Stutter.js (stdin path)	STDIN: "go go " (with trailing space)	Covers stut()->isDelimit(c) with both non-delimiter and delimiter characters; covers delimiter call site stut()->checkDupes(linecnt); covers main()->stut(inFile) where inFile = stdin via argv.length <= 2.	Program prints: Repeated word on line 1: go go	1, 2, 4
T2	node Stutter.js (stdin path)	STDIN: "abc\nabc"	Covers end-of-line call site stut()->checkDupes(linecnt); linecnt reaches line use when duplicate is detected on line 2.	Program prints: Repeated word on line 2: abc abc	3
T3	node Stutter.js "" (empty filename path)	STDIN: "x x " (with trailing space)	Covers main()->stut(inFile) where argv[2] is the empty string and therefore inFile = stdin through the second definition site in main().	Program prints: Repeated word on line 1: x x	5
T4	node Stutter.js sample.txt (valid file path)	sample.txt content: "red red " (with trailing space)	Covers main()->stut(inFile) where inFile is defined from a file stream created by createReadStream(fileName).	Program prints: Repeated word on line 1: red red	6

Conclusion: T1-T4 collectively satisfy All-Coupling-Uses Coverage for the coupling DU-pairs in part (c). The module-entry call to main() does not contribute a coupling DU-pair because no parameter is passed.